

11/04/2025



Aakash

Medical | IIT-JEE | Foundations

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CODE-A



AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

MM : 720

CST - 3

Time : 180 Mins.

Complete Syllabus of Class XI & XII

Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are three parts in the question paper consisting of Physics, Chemistry, Biology which have 45 questions each in Physics & Chemistry and 90 questions in Biology.
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

6 Aakashians scored a perfect 720/720 from Classroom Programs



720 **Mridul M Anand**
720 3 Year Classroom



720 **Ayush Naugraiya**
720 4 Year Classroom



720 **Arghyadeep Dutta**
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720 **Aryan Yadav**
720 1 Year Classroom



720 **Palansha Agarwal**
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720 **Iram Quazi**
720 1 Year Classroom

PHYSICS

1. An electron jumps from orbit $n = 4$ to $n = 3$ in hydrogen atom. Wavelength of the emitted radiation is (R is Rydberg's constant)

- (1) $\frac{144}{7R}$
- (2) $\frac{16}{7R}$
- (3) $\frac{15}{16R}$
- (4) $\frac{8}{9R}$

2. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Fe^{56} nucleus is more stable than U^{235} nucleus.

Reason (R): Binding energy of Fe^{56} nucleus is more than the binding energy of U^{235} .

In the light of the above statements, choose the correct answer from the option given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

3. Boolean expression for the truth table given below is (A and B are inputs and Y is output)

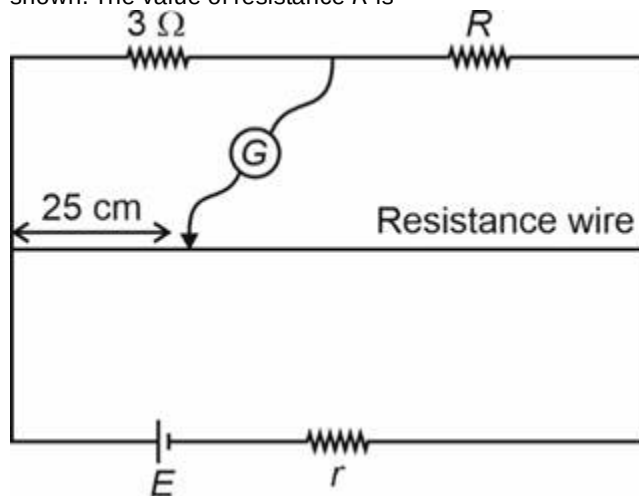
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

- (1) $\overline{A} + \overline{B}$
- (2) $\overline{A} \cdot \overline{B}$
- (3) $A + B$
- (4) $A \cdot B$

4. In a reverse biased diode when the applied voltage changes by 5 V, the current is found to change by $2 \mu\text{A}$. The reverse bias resistance of the diode is

- (1) $10^6 \Omega$
- (2) $2 \times 10^6 \Omega$
- (3) $5 \times 10^6 \Omega$
- (4) $2.5 \times 10^6 \Omega$

5. In an experiment to find the unknown resistance using meter bridge, the null point was found to be at 25 cm as shown. The value of resistance R is



- (1) 1Ω
- (2) 3Ω
- (3) 9Ω
- (4) 27Ω

6. Ultraviolet radiation of 6.1 eV falls on a aluminium surface of work function 4.2 eV. The kinetic energy of the fastest electron emitted is

- (1) $1.9 \times 10^{-19} \text{ J}$
- (2) $1.6 \times 10^{-19} \text{ J}$
- (3) $3.04 \times 10^{-19} \text{ J}$
- (4) $2.5 \times 10^{-19} \text{ J}$

7. Following observations were taken with a vernier calliper while measuring the length of a cylinder: 3.29 cm, 3.28 cm, 3.29 cm, 3.31 cm, 3.28 cm. The most accurate length of the cylinder will be

- (1) 3.28 cm
- (2) 3.29 cm
- (3) 3.32 cm
- (4) 3.21 cm

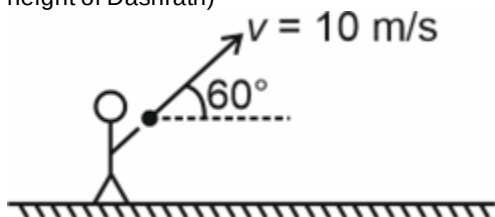
8. A particle covers first half distance of its journey with speed 5 m/s and remaining half distance with speed 10 m/s. The average speed of particle is

- (1) $\frac{15}{2} \text{ m/s}$
- (2) 15 m/s
- (3) $\frac{20}{3} \text{ m/s}$
- (4) $\frac{5}{2} \text{ m/s}$

9. A particle has initial velocity 2 m/s and has constant acceleration such that after 2 s its velocity becomes 3 m/s. The distance travelled by particle in this duration is

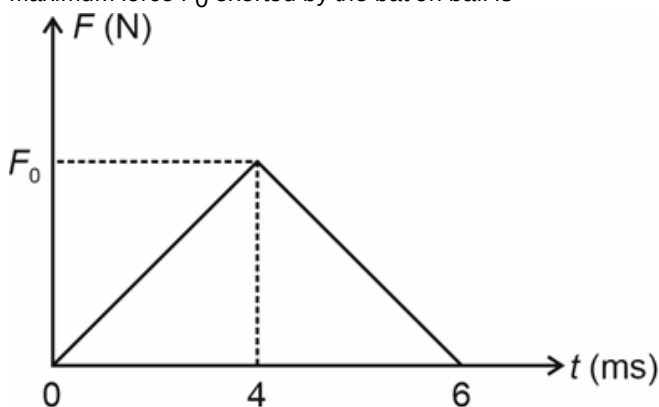
(1) 1 m
(2) 2 m
(3) 5 m
(4) 6 m

10. Dashrath ($m = 60$ kg) is standing on a smooth horizontal surface. If he throws a stone of mass 1 kg with velocity $v = 10$ m/s at angle of 60° with respect to ground then the displacement of Dashrath on the horizontal surface when the stone lands on ground is ($g = 10$ m/s²) (ignore the height of Dashrath)



(1) $\frac{1}{\sqrt{3}}$ m
(2) $\frac{1}{4\sqrt{3}}$ m
(3) $\frac{10}{\sqrt{3}}$ m
(4) $\frac{\sqrt{3}}{20}$ m

11. A ball of mass 100 g is thrown with a speed of 10 m/s. The ball strikes a bat and rebounds along the same line at a speed of 20 m/s. Variation in the interaction force as long as the ball remains in contact with the bat is shown in figure. Maximum force F_0 exerted by the bat on ball is



(1) 4000 N
(2) 1000 N
(3) 500 N
(4) 700 N

12. Arun has $\left(\frac{1}{3}\right)^{\text{rd}}$ of the kinetic energy as that of Raunak when they both run. If Raunak has half the mass of Arun, then the relationship between the speed of Arun v' and speed of Raunak v is

(1) $v' = \frac{v}{\sqrt{6}}$
(2) $v' = \frac{v}{\sqrt{3}}$
(3) $v' = \frac{v}{\sqrt{2}}$
(4) $v' = v$

13. The power of a water pump is 100 kW. If $g = 10$ m/s², then the amount of water it can raise in 2 s to a height of 20 m is

(1) 200 kg
(2) 300 kg
(3) 400 kg
(4) 1000 kg

14. Consider the given statements

Statement A: Measured value by any measuring instrument has some error.

Statement B: A measurement can have more accuracy and less precision and vice-versa.

Statement C: The magnitude of the difference between the true value of the quantity and the individual measured value is called relative error of measurement.

Choose the correct option.

(1) Only statement A is true
(2) Only statement B is true
(3) Both statements A and C are true
(4) Both statements A and B are true

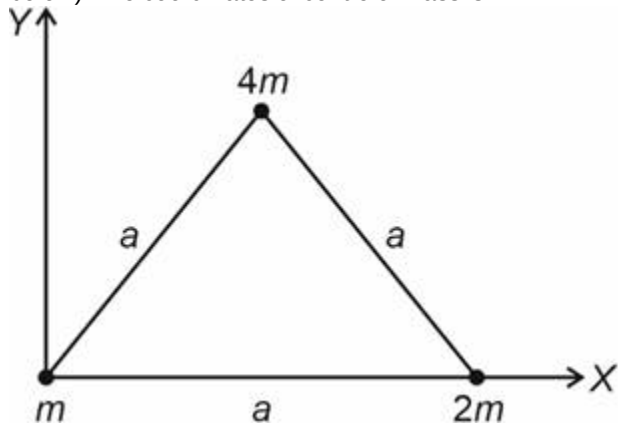
15. Consider the following two statement and tick the correct answer.

Statement A: In projectile motion, horizontal component of velocity of particle remains constant while its acceleration changes continuously.

Statement B: In projectile motion, particle moves such that its velocity and acceleration both changes continuously.

(1) Statement A is correct while statement B is incorrect
(2) Statement A is incorrect while statement B is correct
(3) Both statement A and statement B are correct
(4) Both statement A and statement B are incorrect

16. Three particles of masses m , $2m$ and $4m$ are placed at the corners of an equilateral triangle of side a (as shown below). The coordinates of centre of mass is



- (1) $\left(\frac{a}{4}, \frac{\sqrt{3}a}{4}\right)$
 (2) $\left(\frac{2a}{7}, \frac{2\sqrt{3}a}{7}\right)$
 (3) $\left(\frac{a}{2}, \frac{\sqrt{3}a}{4}\right)$
 (4) $\left(\frac{4a}{7}, \frac{2\sqrt{3}a}{7}\right)$
17. An artificial satellite is moving in a circular orbit of radius r around a planet. Total energy of satellite is E . Energy required to move this satellite into a new orbit of radius $2r$, is

- (1) $\frac{E}{4}$
 (2) $-\frac{E}{4}$
 (3) $\frac{E}{2}$
 (4) $-\frac{E}{2}$

18. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: The mass density of an optically denser medium is greater than that of an optically rarer medium.

Reason R: Optical density is found by the ratio of the speed of light in two media while mass density is mass per unit volume.

In the light of the above statements, choose the correct answer from the option given below.

- (1) Both A and R are true and R is the correct explanation of A
 (2) Both A and R are true but R is not the correct explanation of A
 (3) A is true but R is false
 (4) A is false but R is true

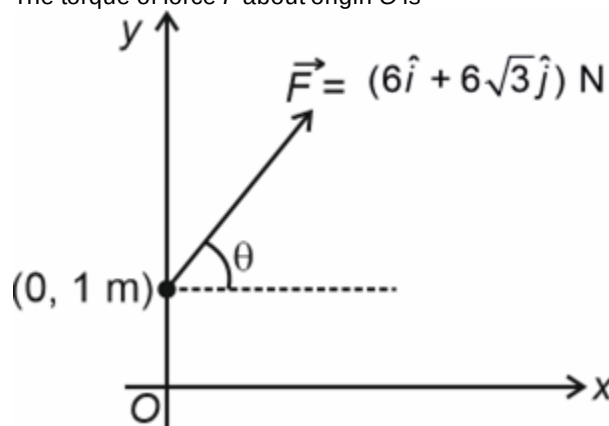
19. In a compound microscope, the focal length of two lenses are 1.25 cm and 12.5 cm. If an object is placed at 5 cm from objective lens and final image is formed at 25 cm from eye-piece lens, the distance between the two lenses is

- (1) 9 cm
 (2) 10 cm
 (3) 11 cm
 (4) 12 cm

20. The factors on which the escape speed from earth depend, is/are

- (1) The height of projection from the earth's surface
 (2) Mass of object
 (3) Mass of earth
 (4) Both (1) and (3)

21. The torque of force F about origin O is



- (1) $(6\hat{k})$ N m
 (2) $(6\sqrt{3}\hat{k})$ N m
 (3) $(-6\hat{k})$ N m
 (4) $(-6\sqrt{3}\hat{k})$ N m
22. Two positive charges q_1 and q_2 having their sum Q are placed at d distance apart. For what values of the charges, the coulomb force between them will be maximum?

- (1) $q_1 = \frac{Q}{4}, q_2 = \frac{3Q}{4}$
 (2) $q_1 = \frac{Q}{3}, q_2 = \frac{2Q}{3}$
 (3) $q_1 = \frac{Q}{2}, q_2 = \frac{Q}{2}$
 (4) $q_1 = \frac{Q}{5}, q_3 = \frac{4Q}{5}$

23. 1000 identical drops of mercury are charged to a potential of 1 V each. They coalesce to form a single drop, the potential of this new drop will be

(1) 100 V
 (2) 10 V
 (3) 1000 V
 (4) 1 V

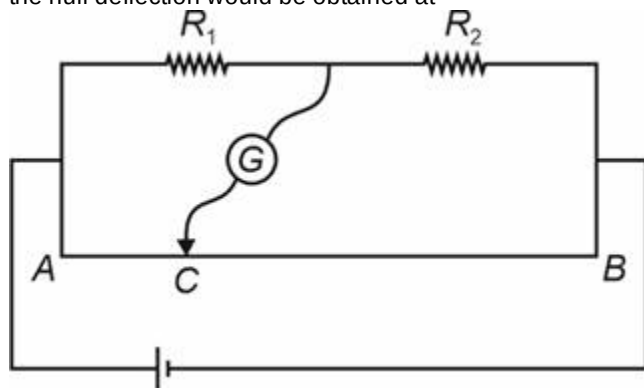
24. Consider the following statements and choose the correct option.

Statement A: Energy is dissipated inside a battery due its internal resistance.

Statement B: In a resistor, current flows from higher potential to lower potential.

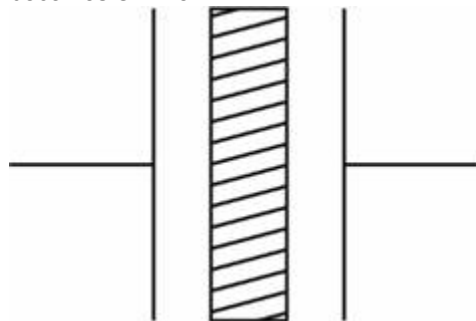
(1) Both the statements are correct
 (2) Both the statements are incorrect
 (3) Only statement A is correct
 (4) Only statement B is correct

25. In the shown arrangement of a meter bridge, when R_1 and R_2 are connected, null deflection is obtained at 40 cm from point A. If R_2 is replaced by another resistance $2R_2$, then the null deflection would be obtained at



(1) AC = 20 cm
 (2) AC = 50 cm
 (3) AC = 30 cm
 (4) AC = 25 cm

26. A parallel plate air capacitor has capacitance C_0 . After it is partially filled with a dielectric as shown its capacitance becomes C' . Then



(1) $C' = C_0$
 (2) $C' < C_0$
 (3) $C' > C_0$
 (4) Insufficient information

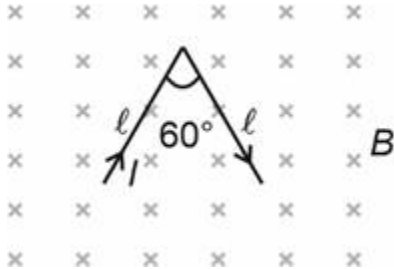
27. A point charge is placed at origin. Assuming potential to be zero at infinity, potential difference at two point A and B is found to be 10 V i.e., $V_A - V_B = 10$ V. Now if the reference at infinity is changed to 10 V, then $V_A - V_B$ will be

(1) 0
 (2) 20 V
 (3) 15 V
 (4) 10 V

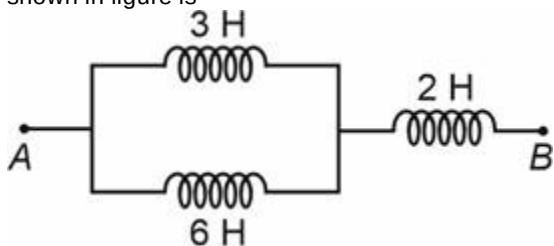
28. A charge particle having mass m and charge q moving with speed v in uniform transverse magnetic field B . The radial acceleration of charge particle will be

(1) $\frac{qB}{mv}$
 (2) $\frac{qBv}{m}$
 (3) $\frac{v^2 qB}{m}$
 (4) $\frac{mv^2}{qB}$

29. A current carrying wire is placed in uniform transverse magnetic field B as shown in figure. The net magnetic force on the wire will be



- (1) $\frac{BIl}{2}$
 (2) $\frac{\sqrt{3}BIl}{2}$
 (3) BIl
 (4) Zero
30. Consider the following statements and choose the correct option.
Statement A: Capacitive reactance is inversely proportional to frequency.
Statement B: Inductive reactance is directly proportional to frequency.
- (1) Both statements A and B are correct
 (2) Both statements A and B are incorrect
 (3) Statement A is correct and B is incorrect
 (4) Statement A is incorrect and B is correct
31. The equivalent inductance between point A and B as shown in figure is



- (1) 11 H
 (2) 4 H
 (3) 2 H
 (4) 6 H
32. The magnetic moment of a magnet having dimensions (10 cm × 5 cm × 4 cm) is 2.5 A m^2 . The intensity of magnetisation of magnet is
- (1) $2.25 \times 10^4 \text{ A/m}$
 (2) $4 \times 10^3 \text{ A/m}$
 (3) $2 \times 10^4 \text{ A/m}$
 (4) $1.25 \times 10^4 \text{ A/m}$

33. Which among the following unit represents henry?

- (1) $\frac{\text{Volt (second)}^2}{\text{Ampere}}$
 (2) $\frac{\text{Volt-second}}{\text{Coulomb}}$
 (3) $\frac{\text{Volt-second}}{\text{Ampere}}$
 (4) $\frac{\text{Volt-second}^3}{\text{Coulomb}}$

34. An ac current flowing in a circuit is given by $i = i_0 \sin \omega t$. The minimum time taken to reach zero to rms value of current is

- (1) $\frac{\pi}{2\omega}$
 (2) $\frac{\pi}{4\omega}$
 (3) $\frac{\pi}{\omega}$
 (4) $\frac{2\pi}{\omega}$

35. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: Strain is a dimensionless quantity.

Reason R: Unit of Young's modulus is same as that of stress.

In the light of the above statements, choose the correct answer from the option given below.

- (1) Both A and R are true and R is the correct explanation of A
 (2) Both A and R are true but R is not the correct explanation of A
 (3) A is true but R is false
 (4) A is false but R is true

36. Equation of SHM of a particle whose amplitude is 0.1 m and frequency is 25 Hz with an initial phase of $\frac{\pi}{4}$ radians is

- (1) $x = 0.1 \sin \left(2\pi t + \frac{\pi}{4} \right)$
 (2) $x = 0.1 \sin \left(2\pi t - \frac{\pi}{4} \right)$
 (3) $x = 0.1 \sin \left(50\pi t + \frac{\pi}{4} \right)$
 (4) $x = 0.1 \sin (50\pi t)$

37. Match list-I with list-II about effect of various factors on speed of sound in a gas.

	List-I (Factor)		List-II (Speed)
a.	With increase in temperature	(i)	Same
b.	With increase in molecular weight of gas keeping temperature constant	(ii)	Increases
c.	With increase in pressure at constant temperature	(iii)	Decreases

- (1) a(ii), b(i), c(iii)
 (2) a(ii), b(iii), c(i)
 (3) a(iii), b(ii), c(i)
 (4) a(i), b(ii), c(ii)

38. Part of EM spectrum which is used to destroy bacteria and sterilizing the surgical instruments

- (1) Infra-red waves
 (2) UV rays
 (3) Microwaves
 (4) Visible light

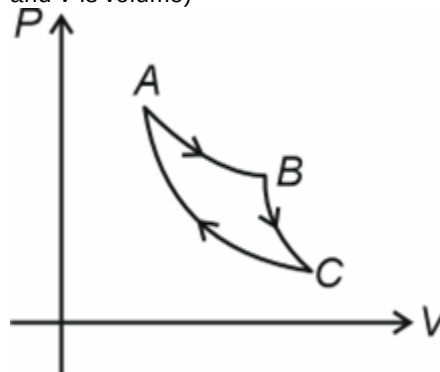
39. Two sources are said to be coherent if they

- (1) Emit same frequency
 (2) Vibrate with a constant phase difference
 (3) Both (1) and (2)
 (4) Vibrate with a changing phase difference

40. In a thermodynamic process, pressure (in Pa) varies as, $P = a + bV$ [where V represents volume in m^3 , a and b are constants]. The work done by gas during expansion from 2 m^3 to 3 m^3 will be

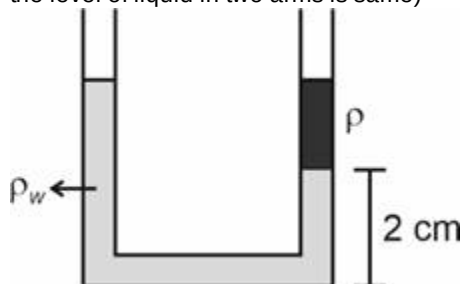
- (1) $\left(\frac{3a}{2} + b\right) \text{ J}$
 (2) $\left(a + \frac{5b}{2}\right) \text{ J}$
 (3) $\left(b + \frac{5a}{2}\right) \text{ J}$
 (4) Zero

41. An ideal gas expands isothermally along AB and does 700 J of work. The gas then expands adiabatically along BC and does 400 J of work. When gas returns to A along CA , it exhausts 100 J of heat to its surrounding. The net heat interaction of gas during the process will be (P is pressure and V is volume)



- (1) Zero
 (2) 700 J
 (3) 600 J
 (4) 400 J

42. The figure represent a U -tube of uniform cross-section filled with two immiscible liquids. One is water with density ρ_w and other liquid is of density ρ . The liquid interface lies 2 cm above base. The relation between ρ and ρ_w is (assume that the level of liquid in two arms is same)



- (1) $\rho = \rho_w$
 (2) $\rho = 1.02 \rho_w$
 (3) $\rho = 1.2 \rho_w$
 (4) Insufficient information

43. If a soap bubble contracts, the pressure inside bubble

- (1) Remains same
 (2) Is equal to atmospheric pressure
 (3) Decreases
 (4) Increases

44. Four particles have velocities 1, 0, 2 and 3 m/s. The root mean square velocity of the particles is
- (1) $\sqrt{3.5}$ m/s
 - (2) $\sqrt{5.3}$ m/s
 - (3) $\sqrt{2.8}$ m/s
 - (4) $\sqrt{4.7}$ m/s
45. Consider the following statements
- A. The temperature at which the liquid and solid state of substance is in thermal equilibrium with each other is called its melting point.
- B. The temperature at which the liquid and the vapour states of the substance coexist is called its boiling point.
- C. The change from solid state to vapour state without passing through liquid state is called sublimation.
- Based on above information pick correct option.
- (1) Only A and B correct
 - (2) Only C and B are correct
 - (3) Only C is incorrect
 - (4) All statements A, B and C are correct

CHEMISTRY

46. Choose the correct order of basic strength for the given compounds
- (1) $\text{Ce}(\text{OH})_3 > \text{Lu}(\text{OH})_3 > \text{Er}(\text{OH})_3 > \text{Gd}(\text{OH})_3$
 - (2) $\text{Gd}(\text{OH})_3 > \text{Er}(\text{OH})_3 > \text{Lu}(\text{OH})_3 > \text{Ce}(\text{OH})_3$
 - (3) $\text{Ce}(\text{OH})_3 > \text{Gd}(\text{OH})_3 > \text{Er}(\text{OH})_3 > \text{Lu}(\text{OH})_3$
 - (4) $\text{Lu}(\text{OH})_3 > \text{Er}(\text{OH})_3 > \text{Gd}(\text{OH})_3 > \text{Ce}(\text{OH})_3$
47. The moles of BaSO_4 formed upon mixing 100 mL of 20.8% (w/V) BaCl_2 solution with 50 mL of 9.8% (w/V) H_2SO_4 solution will be (Molar mass of Ba = 137 g mol⁻¹)
- (1) 0.5
 - (2) 0.05
 - (3) 0.4
 - (4) 0.02
48. At standard conditions, if the change in enthalpy for the following reaction is -109 kJ mol⁻¹
- $$\text{H}_2(\text{g}) + \text{Br}_2(\text{l}) \rightarrow 2\text{HBr}(\text{g})$$
- If bond energy of H_2 and HBr is 435 kJ mol⁻¹ and 368 kJ mol⁻¹ respectively, then is the bond energy in (kJ mol⁻¹) of Br_2 will be
- (1) 736
 - (2) 192
 - (3) 544
 - (4) 276

49. Match list-I with list-II

	List-I (Complexes)		List-II (Spin only magnetic moment)
a.	$[\text{Fe}(\text{CN})_6]^{3-}$	(i)	5.9 BM
b.	$[\text{FeF}_6]^{3-}$	(ii)	0 BM
c.	$[\text{Fe}(\text{CN})_6]^{4-}$	(iii)	4.9 BM
d.	$[\text{CoF}_6]^{3-}$	(iv)	1.7 BM

Choose the correct option among the following.

- (1) a(iv), b(i), c(ii), d(iii)
 - (2) a(i), b(iv), c(iii), d(ii)
 - (3) a(iv), b(ii), c(i), d(iii)
 - (4) a(i), b(iii), c(ii), d(iv)
50. Consider the following statements.
- Statement I:** For irreversible expansion of an ideal gas under isothermal condition, $\Delta U = 0$ but $\Delta S_{\text{total}} \neq 0$.
- Statement II:** For free expansion of an ideal gas under adiabatic conditions, $q = 0$ but $w \neq 0$
- In the light of above statements choose the **correct** option.
- (1) Both statement I and statement II are correct
 - (2) Both statement I and statement II are incorrect
 - (3) Statement I is correct but statement II is incorrect
 - (4) Statement I is incorrect but statement II is correct

51. The correct order for increasing field strength of ligands is

- (1) $\text{SCN}^- < \text{NCS}^- < \text{C}_2\text{O}_4^{2-} < \text{NH}_3$
- (2) $\text{SCN}^- < \text{C}_2\text{O}_4^{2-} < \text{NCS}^- < \text{NH}_3$
- (3) $\text{C}_2\text{O}_4^{2-} < \text{NCS}^- < \text{NH}_3 < \text{SCN}^-$
- (4) $\text{NCS}^- < \text{NH}_3 < \text{C}_2\text{O}_4^{2-} < \text{SCN}^-$

52. Consider the following statements.

- a. Vanadium has highest value of enthalpy of atomisation among all the 3d series elements.
 - b. Cr (VI) in the form of dichromate in acidic medium is a strong oxidising agent.
 - c. Sc^{3+} in aqueous medium is a colourless species.
- Choose the correct statement(s) from the given options.

- (1) a only
- (2) a and b only
- (3) c only
- (4) a, b and c

53. Arrange the following amines in their decreasing order of basic nature in aqueous solution

- (a) Methanamine
 - (b) N-Methylmethanamine
 - (c) N,N-Dimethylmethanamine
- Choose the **correct** option.

- (1) (a) > (b) > (c)
- (2) (c) > (a) > (b)
- (3) (c) > (b) > (a)
- (4) (b) > (a) > (c)

54. Consider the following two statements.

Statement I: Aniline on heating with chloroform and ethanolic potassium hydroxide forms isocyanide.

Statement II: Benzenediazonium chloride reacts with aniline in the presence of acid to form p-Aminoazobenzene (Yellow dye)

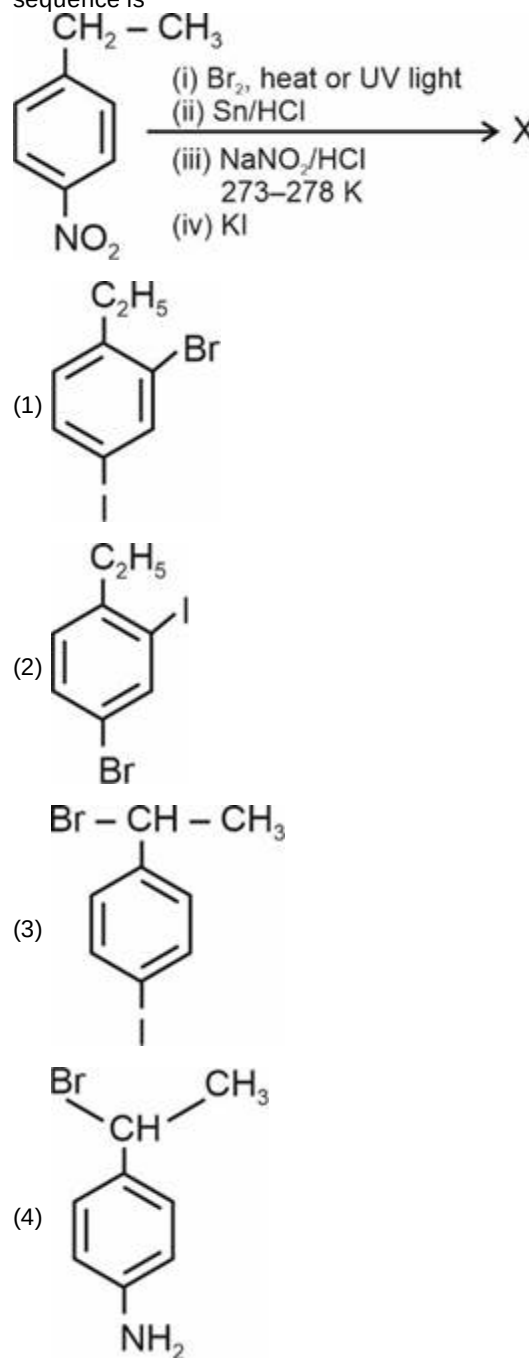
In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

55. Which of the following has highest dipole moment?

- (1) $\text{CH}_3 - \text{F}$
- (2) $\text{CH}_3 - \text{Cl}$
- (3) $\text{CH}_3 - \text{Br}$
- (4) $\text{CH}_3 - \text{I}$

56. The major product X formed in the following reaction sequence is



57. Consider the following two statements.

Statement I: If red blood cells are placed in a solution containing more than 0.9% (w/V) NaCl, the solution is said to be hypertonic.

Statement II: When two isotonic solutions are separated by semipermeable membrane no osmosis occurs between them.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

58. 31g of ethylene glycol is mixed with 750g of water. Calculate the freezing point of the solution. (K_f for water = $1.86 \text{ K kg mol}^{-1}$)

- (1) 1.24°C
- (2) -1.24°C
- (3) 278.24 K
- (4) -269.54 K

59. Given below are two statements one is labelled as assertion (A) and other is labelled as Reason (R)

Assertion (A): Dipole moment of NF_3 is greater than that of NH_3 .

Reason (R): F is more electronegative than H.

In the light of the above statements, choose the correct answer from the options given below

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is false but (R) is true
- (4) (A) is true but (R) is false

60. Identify the number of linear species among the following. SO_2 , I_3^- , CO_2 , H_2O and BeCl_2

- (1) 5
- (2) 2
- (3) 3
- (4) 4

61. The correct number of paramagnetic species among the following is

N_2 , N_2^+ , N_2^- , O_2^- , and O_2^{2-}

- (1) 5
- (2) 2
- (3) 3
- (4) 4

62. Given below are two statements.

Statement I: Time of completion for zero order reaction is twice of its half life period.

Statement II: Inversion of cane sugar is an example of pseudo first order reaction.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

63. If the rate of a reaction doubles when temperature changes from 27°C to 57°C , then approximate value of energy of activation will be

- (1) 4.25 kJ/mol
- (2) 18.96 kJ/mol
- (3) 8.41 kJ/mol
- (4) 12.75 kJ/mol

64. Consider the given compounds

(I) CH_3CHO

(II) HCHO

(III) $(\text{CH}_3)_3\text{CCHO}$

(IV) $\text{C}_6\text{H}_5\text{CHO}$

(V) CH_3COCH_3

Aldol condensation reaction can be observed in

- (1) (I) and (V) only
- (2) (II), (III) and (IV) only
- (3) (I), (II), (IV) and (V) only
- (4) (I), (II), (III) and (IV) only

65. Given below are two statements

Statement I: Carbonate salts give brisk effervescence on reaction with dilute H_2SO_4 .

Statement II: The group reagent for both Al^{3+} and Fe^{3+} is NH_4OH in the presence of NH_4Cl .

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

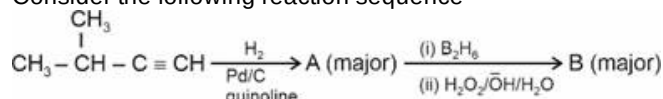
66. Identify the incorrect match for the colour of cations in dry heating test.

	Colour when cold		Cations
a.	Blue	(i)	Cu^{2+}
b.	Green	(ii)	Fe^{2+}
c.	Yellow	(iii)	Zn^{2+}
d.	Pink	(iv)	Co^{2+}

Choose the correct option

- (1) a(i)
- (2) b(ii)
- (3) c(iii)
- (4) d(iv)

67. Consider the following reaction sequence



Major product B is

- (1) $\text{CH}_3 - \underset{\text{CH}_3}{\text{C}}(\text{OH}) - \text{CH}_2 - \text{CH}_3$
- (2) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_2 - \text{CH}_2 - \text{OH}$
- (3) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \underset{\text{OH}}{\text{CH}} - \text{CH}_3$
- (4) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \underset{\text{O}}{\text{C}} - \text{CH}_3$

68. Identify the correct sugar moiety present in DNA and RNA respectively

- (1) α -D-ribose and β -D-2-deoxyribose
- (2) β -D-ribose and β -D-2-deoxyribose
- (3) β -D-2-deoxyribose and β -D-ribose
- (4) α -D-2-deoxyribose and α -D-ribose

69. Consider the following statements regarding cathode rays
- (a) They start from anode and move towards cathode.
 - (b) The characteristics of cathode rays do not depend upon the material of electrodes.
 - (c) The charge to mass ratio of these particle do not depend on the gas present in the cathode ray tube.
- The correct statements are

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a) and (c) only
- (4) (a), (b) and (c)

70. Glucose does not react with

- (1) Br_2 water
- (2) HCN
- (3) NH_2OH
- (4) NaHSO_3

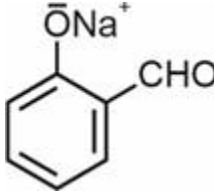
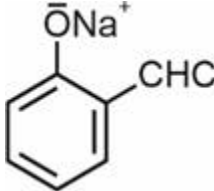
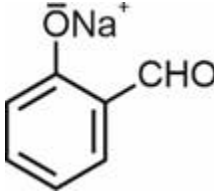
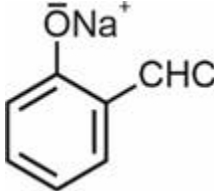
71. Match List-I with List-II

	List-I Reaction		List-II Catalyst
a.	$2\text{CH}_4 + \text{O}_2 \xrightarrow[100\text{ atm}]{523\text{ K}} 2\text{CH}_3\text{OH}$	(i)	Ni
b.	$\text{CH}_4 + \text{H}_2\text{O} \xrightarrow{1273\text{ K}} \text{CO} + 3\text{H}_2$	(ii)	Cu
c.	$\text{CH}_4 + \text{O}_2 \xrightarrow{\Delta} \text{HCHO} + \text{H}_2\text{O}$	(iii)	$(\text{CH}_3\text{COO})_2\text{Mn}$
d.	$2\text{CH}_3\text{CH}_3 + 3\text{O}_2 \xrightarrow{\Delta} 2\text{CH}_3\text{COOH} + 2\text{H}_2\text{O}$	(iv)	Mo_2O_3

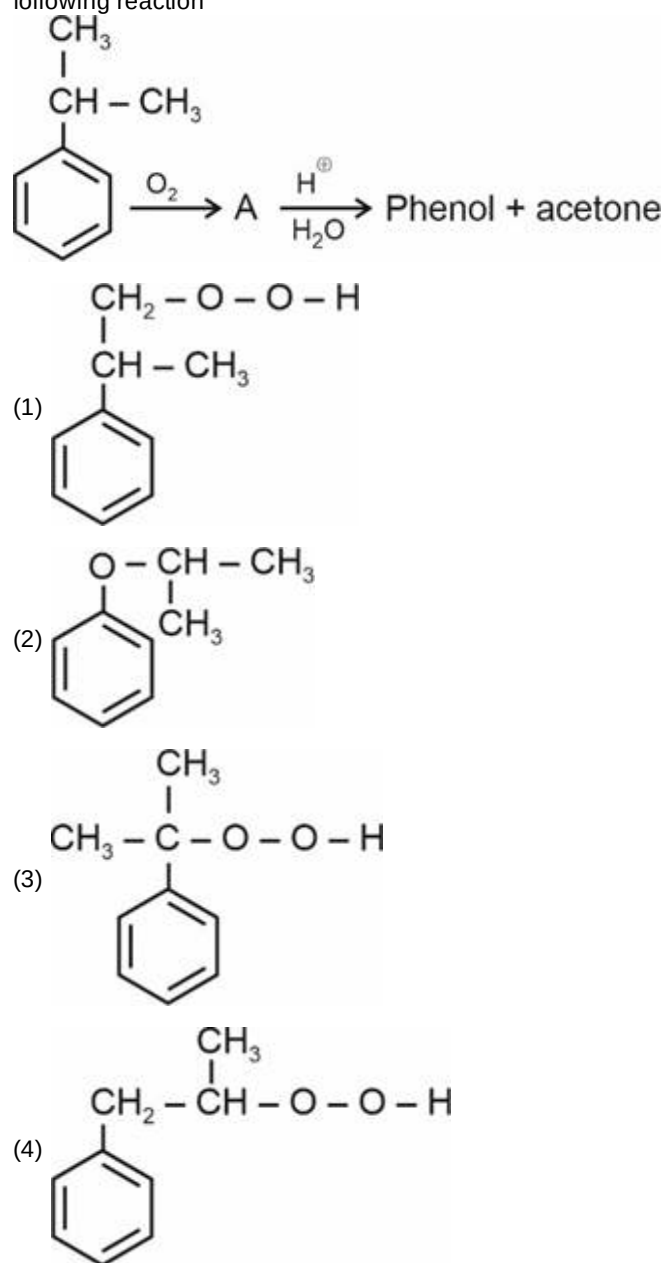
Choose the correct answer from the given options.

- (1) a-(i), b-(ii), c-(iv), d-(iii)
- (2) a-(ii), b-(i), c-(iv), d-(iii)
- (3) a-(ii), b-(i), c-(iii), d-(iv)
- (4) a-(iv), b-(ii), c-(i), d-(iii)

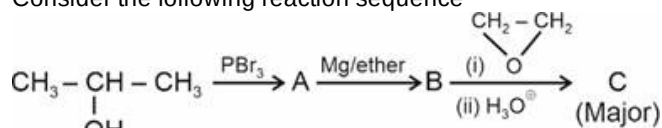
72. Identify the correct intermediate and electrophile formed respectively in the Reimer-Tiemann reaction

- (1)  and $\text{CHO}^\oplus$
- (2)  and $\text{CHCl}_2^\oplus$
- (3)  and :CCl_2
- (4)  and :CCl_2

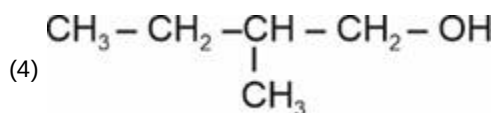
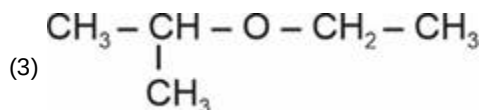
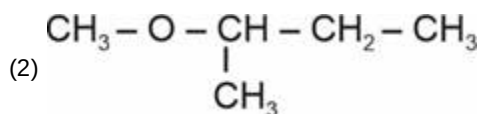
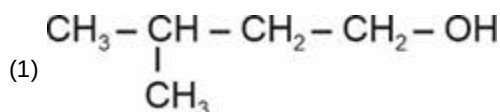
73. Identify correct structure of intermediate compound A in the following reaction



74. Consider the following reaction sequence



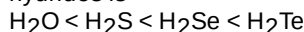
Identify the option with correct structure of major product (C)



75. Given below are two statements

Statement I: Se_2Cl_2 undergoes disproportionation reaction as: $2\text{Se}_2\text{Cl}_2 \rightarrow \text{SeCl}_4 + 3\text{Se}$

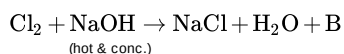
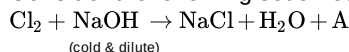
Statement II: The correct order for boiling point of Group 16 hydrides is



In the light of above statements, choose the correct option.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

76. Consider the following set of reactions.



Products A and B respectively are

- (1) NaClO_3 and NaClO_4
- (2) NaOCl and NaClO_4
- (3) NaClO_3 and NaOCl
- (4) NaOCl and NaClO_3

77. White phosphorus when heated with concentrated NaOH solution in an inert atmosphere of CO_2 produces

- (1) PH_3 and NaH_2PO_3
- (2) PH_3 and NaH_2PO_2
- (3) P_2H_4 and NaH_2PO_3
- (4) PH_3 and NaH_2PO_4

78. Given below are two statements.

Statement I: Negative electron gain enthalpy of oxygen is more than that of negative electron gain enthalpy of sulphur.

Statement II: First ionization enthalpy of sodium is more than lithium.

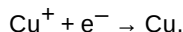
In the light of above statements choose the correct option.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

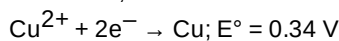
79. The correct order of ionic radii of the given species is

- (1) $\text{Na}^+ > \text{Al}^{3+} > \text{Mg}^{2+} > \text{Si}^{4+}$
- (2) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Si}^{4+} > \text{Na}^+$
- (3) $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+} > \text{Si}^{4+}$
- (4) $\text{Si}^{4+} > \text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+$

80. Standard electrode potential for the following half cell reaction will be



Given that,



Choose the correct option.

- (1) 1.12 V
- (2) 0.52 V
- (3) 0.26 V
- (4) 0.13 V

81. Consider the following limiting molar conductances

$$\Lambda_m^0(\text{HCl}) = x \text{ S cm}^2 \text{ mol}^{-1}$$

$$\Lambda_m^0(\text{CH}_3\text{COOK}) = y \text{ S cm}^2 \text{ mol}^{-1}$$

$$\Lambda_m^0(\text{KCl}) = z \text{ S cm}^2 \text{ mol}^{-1}$$

Λ_m^0 (in $\text{S cm}^2 \text{ mol}^{-1}$) for CH_3COOH will be

- (1) $x - y + z$
- (2) $z - x - y$
- (3) $y + x - z$
- (4) $y - x - z$

82. If 2.88 g of sodium benzoate is present in 1000 mL of an aqueous solution, then pH of the solution will be (pKa of benzoic acid = 4.2, molar mass of sodium benzoate = 144 g mol^{-1})

- (1) 13.1
- (2) 6.27
- (3) 8.25
- (4) 10.2

83. The solubility of AgCl(s) with solubility product 1.6×10^{-10} in 0.2 M NaCl solution will be

(1) 8×10^{-9}
 (2) 6.4×10^{-10}
 (3) 8×10^{-10}
 (4) 1.6×10^{-9}

84. In which of the given compounds the oxidation state of sulphur is found to be the least?

(1) S_8
 (2) H_2SO_4
 (3) SO_2
 (4) H_2S

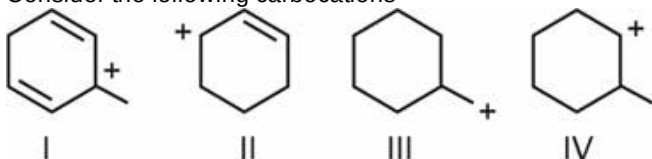
85. Match the Column I with Column II

	Column I (Compound)		Column II (Number of σ bonds)
a.	Benzene	(i)	16
b.	Anisole	(ii)	15
c.	Aniline	(iii)	14
d.	Toluene	(iv)	12

Choose the most appropriate answer the options given below.

(1) a(iii), b(ii), c(iv), d(i)
 (2) a(iv), b(i), c(iii), d(ii)
 (3) a(iv), b(i), c(ii), d(iii)
 (4) a(iii), b(iv), c(i), d(ii)

86. Consider the following carbocations



The correct order of stability of the carbocations is

(1) $\text{I} > \text{IV} > \text{II} > \text{III}$
 (2) $\text{I} > \text{II} > \text{IV} > \text{III}$
 (3) $\text{II} > \text{I} > \text{III} > \text{IV}$
 (4) $\text{II} > \text{IV} > \text{I} > \text{III}$

87. Heterocyclic compound(s) among the following is/are

(i) Pyrrole
 (ii) Furan
 (iii) Pyridine
 (iv) Tropone
 Choose the correct option

(1) (i) and (iv) only
 (2) (i) and (iii) only
 (3) (i), (ii) and (iii) only
 (4) (iv) only

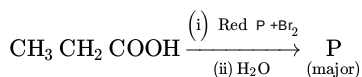
88. Which of the given species is least likely to exist?

(1) $[\text{SiF}_6]^{2-}$
 (2) CCl_4
 (3) BF_4^-
 (4) $[\text{SiCl}_6]^{2-}$

89. The wavenumber for the longest wavelength transition in the Balmer series of atomic hydrogen is (R_H = Rydberg constant of hydrogen atom)

(1) $\frac{3}{4} R_H$
 (2) $\frac{5}{36} R_H$
 (3) $\frac{4}{3R_H}$
 (4) $\frac{36}{4R_H}$

- 90.



Product 'P' is

(1) $\text{CH}_3\text{CH}_2\text{COBr}$
 (2) $\text{CH}_3\text{CHBrCOOH}$
 (3) $\text{CH}_3\text{CH}_2\text{Br}$
 (4) $\text{CH}_2\text{BrCH}_2\text{COOH}$

91. Which of the following statements is **incorrect** regarding bryophytes?
- (1) The main plant body is diploid.
 - (2) They are dependent on water for sexual reproduction.
 - (3) They lack true roots, stems or leaves
 - (4) Some cells of the sporophyte undergo reduction division to produce haploid spores.
92. Mark the following statements as true (T) or false (F) and select the **correct** option.
- (a) In *Marchantia*, the thallus is dorsiventral and closely appressed to the substrate.
 - (b) In pteridophytes, the main plant body is differentiated into true roots, stem and leaves.
 - (c) Prothallus require cool, damp, shady place to grow.
 - (d) In majority of pteridophytes, two kinds of spores are produced.
- (1) a(T), b(F), c(T), d(T)
 - (2) a(F), b(F), c(T), d(F)
 - (3) a(T), b(T), c(T), d(F)
 - (4) a(T), b(T), c(F), d(F)
93. Cyclosporin A, a bioactive molecule is produced by
- (1) *Monascus purpureus*
 - (2) *Trichoderma polysporum*
 - (3) *Aspergillus niger*
 - (4) *Streptococcus*
94. In which of the following interactions, both the organisms are adversely affected?
- (1) Amensalism
 - (2) Commensalism
 - (3) Mutualism
 - (4) Competition
95. In general, loss of biodiversity in a region may lead to
- (1) Increase in plant production
 - (2) Lowered resistance to environmental perturbations
 - (3) Increased resilience to man made disturbances
 - (4) Decreased variability in pest and disease cycles
96. The Nile Perch introduced into Lake Victoria in east Africa led eventually to the extinction of
- (1) Cichlid fish
 - (2) *Clarias gariepinus*
 - (3) Catfishes
 - (4) Steller's sea cow
97. Which of the given is similar in structure to amylopectin and glycogen?
- (1) Starch
 - (2) Mannitol
 - (3) Laminarin
 - (4) Floridean starch
98. The effluent from the primary settling tank is
- (1) Biologically treated by anaerobic bacteria to lower down its polluting potential.
 - (2) Passed into large aeration tanks where it is constantly agitated mechanically.
 - (3) Generally released into natural water bodies like rivers and streams.
 - (4) Highly polluted and have low biochemical oxygen demand.
99. Identify the **correctly** matched pair.
- (1) Oyster – Produce small number of large sized offsprings
 - (2) Birds - Regulators
 - (3) *Chathamalus* – dominates intertidal area
 - (4) *Calotropis* – Produce strychnine
100. During DNA replication, the discontinuously synthesized fragments are joined by an enzyme called
- (1) Topoisomerase
 - (2) DNA ligase
 - (3) Primase
 - (4) DNA polymerase
101. Select the **correctly** matched pair
- (1) tRNA - has a guanine residue at its 5' end
 - (2) RNA polymerase II - transcribes small nuclear RNA
 - (3) Ribozyme - 28S rRNA in *E.coli*
 - (4) RNA polymerase I – transcribes 5S rRNA
102. Chief water conducting tissue in gymnosperms lacks
- (1) Xylem fibres
 - (2) Vessels
 - (3) Xylem parenchyma
 - (4) Tracheids

103. Match List-I with List-II

	List-I		List-II
a.	Bast fibres	(i)	One of the components of stomatal apparatus
b.	Companion cells	(ii)	Make endodermis impervious to water
c.	Casparian strips	(iii)	Generally absent in primary phloem
d.	Subsidiary cells	(iv)	Maintain pressure gradient in sieve tubes

Choose the **correct** option.

- (1) a(iii), b(iv), c(ii), d(i)
- (2) a(i), b(ii), c(iii), d(iv)
- (3) a(iii), b(i), c(iv), d(ii)
- (4) a(ii), b(iii), c(i), d(iv)

104. Select the **correct** statement(s) w.r.t. satellite DNA.

- (a) It is classified on the basis of base composition, length of segment and number of repetitive units
- (b) Its sequences normally code for proteins needed in cell cycle
- (c) It forms a small portion of human genome
- (d) It does not shed light on chromosomal structure and its dynamics

Choose the **correct** option.

- (1) (b), (c) and (d) only
- (2) (a) and (b) only
- (3) (a) and (c) only
- (4) (a) only

105. Read the following Assertion (A) and Reason (R) and choose the **correct** option.

Assertion (A): Ribonucleic acid is labile and easily degradable as compared to DNA.

Reason (R): Presence of free 2'-OH group at every nucleotide in RNA, is a reactive group and makes it more labile and easily degradable.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

106. Hepatic portal vein carries blood from

- (1) Intestine to liver
- (2) Lower parts of the body to kidney
- (3) Intestine to kidney
- (4) Liver to intestine

107. Nicotinamide adenine dinucleotide (NAD) and NADP contain the vitamin niacin. Here,

- (1) Niacin acts as the prosthetic group
- (2) NAD acts as a co-enzyme
- (3) NADP is tightly bound to the apoenzyme
- (4) Niacin forms strong coordination bonds with side chains at the active site

108. Choose the biomacromolecule which forms the least per cent of the total cellular mass.

- (1) Protein
- (2) Lipid
- (3) Nucleic acid
- (4) Carbohydrate

109. A student was analysing the structure of the respiratory pigment 'X' which is responsible for carrying O₂ in healthy adult humans. He found that 'X'

- (1) Possesses two subunits of α -type and two subunits of β -type
- (2) Exhibits only tertiary structure in which the protein thread is folded in the form of a helix
- (3) Consists of covalent, hydrogen, disulphide as well as phosphodiester bonds
- (4) Possesses only one subunit of α -type

110. All of the following are true w.r.t curcumin, **except**

- (1) It is a secondary metabolite
- (2) Belongs to the same category as vinblastin
- (3) It is a drug
- (4) It is a polymeric substance

111. Choose the odd one w.r.t secretions of exocrine glands.

- (1) Mucus
- (2) Earwax
- (3) Secretin
- (4) Pepsin

112. Read the statements given below w.r.t humans:

Statement A: Pons consists of fibre tracts that interconnect different regions of the brain.

Statement B: The efferent fibres transmit regulatory impulses from the CNS to the concerned peripheral tissues/organs.

Select the **correct** option.

- (1) Both statements A and B are correct
- (2) Both statements A and B are incorrect
- (3) Only statement A is correct
- (4) Only statement B is correct

113. Select the **incorrect** match w.r.t. rDNA technology.

- (1) Cloning - Making multiple identical copies of any template DNA
- (2) Transformation - Procedure through which a piece of DNA is introduced in a host bacterium
- (3) Recombinant protein - Any protein encoding gene expressed in a homologous host
- (4) Elution - After gel electrophoresis, the separated bands of DNA are cut out from the agarose gel and extracted from the gel piece

114. All of the following are true w.r.t *Agrobacterium tumefaciens*, **except**

- (1) Acts as a natural genetic engineer
- (2) Always induces tumour formation in animal cells
- (3) Can be disarmed to carry the gene of interest in plants
- (4) Contains pathogenic T-DNA in its plasmid

115. Select the **incorrect** option w.r.t. the steps involved in the post-biosynthetic stage of RDT.

- (1) Quality control testing
- (2) Formulation of products with suitable preservatives
- (3) Separation and purification of the desired product
- (4) Culturing the host cells with rDNA in a medium at large scale

116. State true (T) or false (F) for the given statements.

- A. In an isobilateral leaf, sub-stomatal cavity is present below the stoma of the abaxial epidermis.
 B. Sclereids are present in the seed coats of pea and beans but are absent in the leaves of tea.
 Choose the **correct** option.

- (1) A(T), B(T)
- (2) A(F), B(F)
- (3) A(T), B(F)
- (4) A(F), B(T)

117. In prokaryotes, DNA

- (1) Is positively charged
- (2) Is found in coiled stage, maintained only by acidic proteins
- (3) Has single origin of replication
- (4) Is devoid of 5-methyl uracil

118. Which of the following is **odd** w.r.t. the functions of the endomembrane system?

- (1) Digestion of biomolecules
- (2) Formation of lysosomes
- (3) Formation of glycoproteins & glycolipids
- (4) Oxidative phosphorylation

119. Read the following statements and state them as True (T) or False (F) and choose the **correct** option.

- a. Neutral solutes may move across the membrane by the process of simple diffusion.
- b. Secondary constriction and satellite are non-staining region of certain chromosomes.
- c. Inner membrane of mitochondria is selectively permeable and is rich in cardiolipins
- d. In centrioles, peripheral doublets are connected to each other.

- (1) a(T), b(F), c(T), d(F)
- (2) a(T), b(T), c(F), d(F)
- (3) a(F), b(F), c(T), d(T)
- (4) a(F), b(T), c(F), d(T)

120. Which of the following events occurs during the G₁ phase of cell cycle?

- (1) DNA Replication
- (2) Tubulin protein synthesis
- (3) Duplication of centriole
- (4) Duplication of majority of cell organelles

121. Find the **odd** one w.r.t. the zygotene stage of cell cycle.

- (1) Pairing of homologous chromosomes
- (2) Formation of synaptonemal complex
- (3) Crossing over
- (4) Bivalent formation

122. Observe the events given below.

- (a) ATP synthesis
- (b) DNA duplication
- (c) Histone synthesis
- (d) Organelle duplication
- (e) Karyokinesis

Select the events that occurs only during resting phase of cell cycle.

- (1) b and e only
- (2) b, c and d only
- (3) a, c and e only
- (4) d and e only

123. Find the **incorrect** match.

- (1) Slime layer - Cell envelope
- (2) Inclusion bodies - Lie freely in the cytoplasm
- (3) Tonoplast – Envelops vacuole
- (4) Ribosomes – Composed of deoxyribonucleic acid and proteins

124. A secondary consumer in an ecosystem

- (1) Is a herbivorous animal
- (2) Occupies second trophic level
- (3) Is a primary carnivore
- (4) Occupies the base of an ecological pyramid

- 125.**How many molecules of ATP are obtained only from ETS, if one molecule of pyruvic acid is completely oxidised?
- 11
 - 14
 - 12
 - 18
- 126.**Identify the **incorrect** statement(s) from the following and choose the **correct** option.
- Non-cyclic photophosphorylation results in the synthesis of ATP only
 - Stroma lamellae lack PS I as well as NADP reductase enzyme.
 - In cyclic photophosphorylation, PS-I is functional.
- (i) only
 - (ii) and (iii) only
 - (i) and (ii) only
 - (iii) only
- 127.**During Calvin cycle, formation of two molecules of a 3C compound is due to
- Carboxylase activity of RuBisCO
 - Carboxylase activity of PEPcase
 - Oxygenase activity of PEPcase
 - Oxygenase activity of RuBisCO
- 128.**Consider the following statements and select the **correct** option.
- Statement A:** NADH synthesized in glycolysis is transferred into the mitochondria to undergo oxidative phosphorylation.
- Statement B:** Complete oxidation of two FADH_2 molecules leads to the formation of two ATP molecules.
- Only statement A is correct
 - Only statement B is correct
 - Both the statements A and B are correct
 - Both the statements A and B are incorrect
- 129.**Who identified and crystallised the cytokinesis promoting active substance, kinetin?
- F. Skoog
 - H.H. Cousins
 - Miller et al.
 - Darwin
- 130.**Identify the **odd** one w.r.t. the cells in the meristematic phase of growth.
- High respiratory rate
 - Dense protoplasm
 - Secondary cell wall deposition
 - Abundant plasmodesmata connections

- 131.**At what temperature, the heat shock treatment is given that enables the bacteria to take up the rDNA during bacterial transformation?
- 52°C
 - 62°C
 - 42°C
 - 72°C
- 132.**The somatic neural system will relay impulses from CNS to
- Gall bladder
 - Aorta
 - Small intestine
 - Biceps
- 133.**Comprehend the statements given below:
- Traditional hybridisation procedures used in plant and animal breeding, very often lead to inclusion and multiplication of undesirable genes along with the desired genes.
 - The palindrome in DNA is a sequence of base pairs that reads the same on the two strands when orientation of reading is kept the same.
 - The most commonly used matrix is agarose which is a natural monomer extracted from sea weeds.
 - The normal *E. coli* carry resistance against many antibiotics.
- Select the option that represents only the **correct** set of statements.
- (b) and (d)
 - (c) and (d)
 - (b) and (c)
 - (a) and (b)

- 134.**Match the columns w.r.t. joints in human body.

	Column I		Column II
a.	Pivot joint	(i)	In between bones of cranium
b.	Saddle joint	(ii)	Between humerus and pectoral girdle
c.	Ball and socket joint	(iii)	Between carpal and metacarpal of thumb
d.	Fibrous joint	(iv)	Between atlas and axis

Choose the **correct** option.

- a(iv), b(iii), c(i), d(ii)
- a(iv), b(i), c(ii), d(iii)
- a(iii), b(ii), c(i), d(iv)
- a(iv), b(iii), c(ii), d(i)

135.Read the following statements:

- (a) All chordates have a closed circulatory system.
 - (b) All vertebrates are characterised by presence of a closed circulatory system.
 - (c) In fishes, heart pumps out oxygenated blood.
- Choose the **correct** option.

- (1) (a), (b) and (c) are correct
- (2) Only (a) and (b) are correct
- (3) Only (a) and (c) are correct
- (4) Only (b) is correct

136.A single U-shaped bone called hyoid is present at the

- (1) Upper roof of the buccal cavity
- (2) Base of the buccal cavity
- (3) Lateral part of cranium
- (4) Superior part of cranium

137.The type of movement exhibited by human spermatozoa, also helps in

- (1) Movement of food in *Paramoecium*
- (2) Locomotion in *Hydra*
- (3) Movement of macrophages
- (4) Maintenance of water current in the water canal system of *Sycon*

138.Assertion (A): The intercellular material of cartilage is pliable.

Reason (R): The ground substance is rich in chondroitin salts.

In the light of above statements, choose the correct option.

- (1) Both (A) and (R) are true and (R) correctly explains (A)
- (2) Both (A) and (R) are true but (R) does not explain (A) correctly
- (3) (A) is true, (R) is false
- (4) Both (A) and (R) are false

139.All of the following are **correct** w.r.t. WBCs in humans, **except**

- (1) Formed in red bone marrow in adults
- (2) Neutrophils are the most abundant type of WBCs
- (3) All of them are nucleated
- (4) Their number averages around $6000-8000 \text{ cm}^{-3}$ of blood

140.Which of the following structures is made up of special tissue that undergoes erection to facilitate insemination?

- (1) Scrotum
- (2) Penis
- (3) Clitoris
- (4) Foreskin

141.The structures named 'X' leave the testis and open into a tubular structure located along the posterior surface of each testis. Here, 'X' are

- (1) Rete testis
- (2) Vas deferens
- (3) Vasa efferentia
- (4) Epididymis

142.Select the odd one w.r.t. the sensory structures present in cockroaches.

- (1) Pronotum
- (2) Antennae
- (3) Ocellus
- (4) Anal cerci

143.Choose the **correct** option to complete the analogy w.r.t. humans.

Female sex accessory duct : Vagina :: Male sex accessory duct : _____.

- (1) Penis
- (2) Seminal vesicle
- (3) Vas deferens
- (4) Ureter

144.Choose the common mode of respiration in adult frog on land as well as in aquatic habitats from the options given below.

- (1) Branchial
- (2) Buccopharyngeal
- (3) Pulmonary
- (4) Cutaneous

145.How many of the following statements is/are correct w.r.t. parturition in human?

- (a) The process of delivery of embryo is called parturition.
 - (b) Childbirth is induced exclusively by endocrine mechanism.
 - (c) Foetal ejection reflex triggers the release of oxytocin from the maternal pituitary.
 - (d) The whole placenta degenerates within uterus soon after the baby is delivered.
- Choose the **correct** option.

- (1) Two
- (2) Three
- (3) Four
- (4) One

146.Select the odd one w.r.t. the secretion of adrenal cortex.

- (1) Cortisol
- (2) Aldosterone
- (3) Adrenaline
- (4) Androgenic steroid

147. Select the **incorrect** option w.r.t. MALT.

- (1) Stands for mucosa-associated lymphoid tissue
- (2) Is a primary lymphoid organ
- (3) Constitutes about 50% of the lymphoid tissue in humans
- (4) Located within the lining of the respiratory, digestive and urogenital tracts.

148. Usually, the time-lag between the infection and appearance of AIDS symptoms is of

- (1) 10-15 days
- (2) 1-2 months
- (3) 5-10 days
- (4) 5-10 years

149. Match Column I with Column II and select the **correct** option

	Column I		Column II
a.	Aldosterone	(i)	Acts on uterine wall muscles
b.	PTH	(ii)	Stimulates the reabsorption of water and Na^+ by distal tubules
c.	TCT	(iii)	Stimulates the reabsorption of Ca^{2+} by renal tubule
d.	Oxytocin	(iv)	Reduces blood calcium levels

- (1) a(iii), b(iv), c(ii), d(i)
- (2) a(ii), b(iii), c(iv), d(i)
- (3) a(iii), b(ii), c(i), d(iv)
- (4) a(ii), b(iv), c(iii), d(i)

150. The correct set of diseases which can be controlled by introducing fishes in ponds that feed on mosquito larvae are

- (1) Ascariasis, Dengue
- (2) Malaria, Amoebiasis
- (3) Ascariasis, Filariasis
- (4) Filariasis, Dengue

151. Select the odd one with respect to Australian marsupials.

- (1) Spotted cuscus
- (2) Lemur
- (3) Flying phalanger
- (4) Tasmanian wolf

152. When more than one adaptive radiations appeared to have occurred in an isolated geographical area representing different habitats, one can call this

- (1) Saltation
- (2) Convergent evolution
- (3) Mutation
- (4) Genetic drift

153. Animals which are devoid of scales and paired fins and whose adults are marine while larvae are fresh water forms, belong to one of the taxon under

- (1) Division Gnathostomata
- (2) Super class Pisces
- (3) Class Amphibia
- (4) Subphylum Vertebrata

154. Select the **incorrect** statement w.r.t. the cortical nephrons found in humans.

- (1) The loop of Henle present in them is too short and extends very little into the medulla.
- (2) In the kidney, the majority of the nephrons are cortical nephrons.
- (3) The vasa recta is absent or highly reduced in them.
- (4) Peritubular capillaries associated with them emerge from the afferent arteriole.

155. Choose the **correct** match.

(1)	Renal corpuscle	–	A tuft of capillaries formed by the efferent arteriole
(2)	Bowman's capsule	–	Formed by glomerulus along with Malpighian body
(3)	Columns of Bertini	–	Extensions of cortex in between the medullary pyramids
(4)	Renal pelvis	–	Inner concave surface of the kidney

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

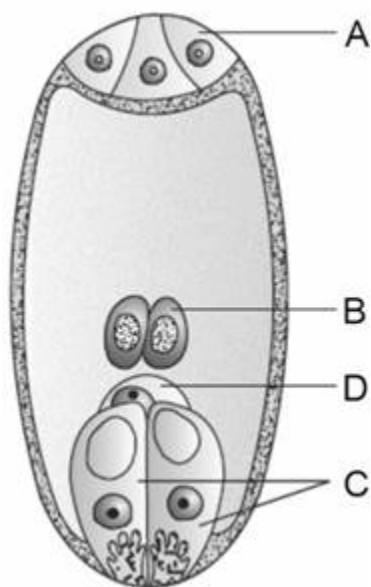
156. Bud pollination is observed in

- (1) Pea
- (2) Maize
- (3) Sugarcane
- (4) *Vallisneria*

157. Which of the following is an **incorrect** statement?

- (1) Production of unisexual flowers is an outbreeding device developed by the plants
- (2) Pollination guarantees the transfer of the right type of pollen on the stigma of a particular plant species
- (3) Dioecy prevents both autogamy and geitonogamy
- (4) Pollination by water is limited to about 30 genera

158. Observe the figure given below.



Choose the **correct** option from the following for 'A', 'B', 'C', 'D'.

	A	B	C	D
(1)	Synergids	Antipodals	Egg	Central cell
(2)	Synergids	Polar nuclei	Antipodals	Egg
(3)	Antipodals	Egg	Polar nuclei	Synergids
(4)	Antipodals	Polar nuclei	Synergids	Egg

- (1) (1)
(2) (2)
(3) (3)
(4) (4)

159. In a typical test cross,

- (1) An organism showing dominant phenotype is crossed with the recessive parent.
(2) An organism showing recessive phenotype is crossed with the heterozygous parent.
(3) Phenotypic and genotypic ratio are always different.
(4) No dominant individuals are obtained in the progeny.

160. Which of the following is the **odd** one w.r.t. Klinefelter's syndrome?

- (1) 44+XXY chromosome complement
(2) Sterile male
(3) Gynaecomastia
(4) Short stature

161. Mark the **correct** ratio obtained for round yellow, wrinkled and green seeds in F_2 generation in the dihybrid cross that was conducted by Mendel.

- (1) 3 : 1
(2) 9 : 1
(3) 1 : 3
(4) 1 : 1

162. Following are two statements, labelled as Assertion (A) and Reason (R).

Assertion (A): A daughter will not normally be colour blind, unless her mother is a carrier and her father is colour blind.

Reason (R): Colour blindness is an X-linked recessive disorder.

In the light of the above statements, identify the **correct** option from the following.

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

163. Read the following statements and select the **correct** option.

Statement-A: Characterisation, identification, classification and nomenclature are the processes that are basic to taxonomy.

Statement-B: Classification is the process by which anything is grouped into convenient categories based on some easily observable characters.

- (1) Only statement A is correct
(2) Only statement B is correct
(3) Both the statements A and B are correct
(4) Both the statements A and B are incorrect

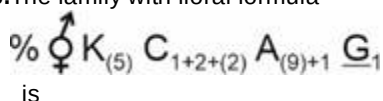
164. Which of the following statements is **not** true w.r.t. Archaeobacteria?

- (1) They have introns in their genetic sequence
(2) They differ from other bacteria in having a different cell wall structure.
(3) They are found in extreme conditions
(4) Their cell membrane contains branched chain lipids which increase membrane fluidity

165. Kingdom fungi is classified into various classes on the basis of all, **except**

- (1) Morphology of mycelium
(2) Fruiting bodies
(3) Mode of spore formation
(4) Reserve food material

166. The family with floral formula



- is
- (1) Fabaceae
 - (2) Solanaceae
 - (3) Liliaceae
 - (4) Brassicaceae

167. Read the following statements (A-D)

- A. Aggregate fruits are formed from polycarpellary, apocarpous ovary.
 B. Composite fruits develop from entire inflorescence.
 C. Micropyle is a scar on the seed coat through which the developing seeds were attached to the fruit.
 D. Middle layer of pericarp in coconut is used in making coir.

The **correct** ones are

- (1) Only A, B and C
- (2) Only B and C
- (3) Only C and D
- (4) Only A, B and D

168. Identify the **incorrectly** matched pair w.r.t. symbols used in floral formula.

(1)	$\overline{\text{C}} \text{ A}$	-	Epipetalous stamen
(2)	$\text{P} \text{ A}$	-	Epiphyllous stamen
(3)	$\text{K}_{(n)}$	-	Polysepalous condition
(4)	$\underline{\text{G}}_n$	-	Superior ovary

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

169. Identify the following statements as true(T) or false(F) and choose the option accordingly.

- a. T.O. Diener discovered a new infectious agent that was smaller than viruses and caused potato spindle tuber disease.
 b. Viruses are obligate parasites and each contain both RNA and DNA.
 c. Lichens are very good pollution indicators as they grow easily in polluted areas.
 d. The most notable disease caused by prions is bovine spongiform encephalopathy, commonly called mad cow disease.

- (1) a(T), b(T), c(F), d(T)
- (2) a(T), b(F), c(F), d(T)
- (3) a(F), b(F), c(T), d(T)
- (4) a(T), b(F), c(T), d(F)

170. How many of the following food chains form upright pyramid of number?

- a. Tree → Birds → Parasite
- b. Seeds → Sparrows → Cats
- c. Grasses → Herbivores → Lions
- d. Shrubs → Goat → Tiger

- (1) Two
- (2) Three
- (3) Four
- (4) Five

171. Darwin's theory of evolution is based on two key concepts. Identify them among the following and select the **correct** option.

- (i) Saltation
- (ii) Genetic drift
- (iii) Physical fitness
- (iv) Branching descent
- (v) Natural selection
- (vi) Mutation

- (1) (i) and (vi)
- (2) (iii) and (vi)
- (3) (ii) and (v)
- (4) (iv) and (v)

172. Read the following features:

- (a) Stinging capsules are present in specialised cells on tentacles
- (b) Exhibits umbrella-shaped body form
- (c) Sessile in nature and have radial symmetry
- (d) Has a gastrovascular cavity in which extracellular digestion occurs

How many of the above given features is/are **correct** for sea anemone?

- (1) One
- (2) Four
- (3) Three
- (4) Two

173. The first clinical gene therapy was given to a girl in the year 'X'. Choose the event which occurred after the year 'X'.

- (1) An American company prepared genetically engineered human insulin
- (2) The first transgenic cow produced human protein-enriched milk.
- (3) First recombinant DNA was made.
- (4) Enzymes responsible for restricting the growth of bacteriophage in *E. coli* were discovered

174. If a new biotech company wants a patent for their recently-made product, then the product should be

- (1) Already available in the market
- (2) Already identified by indigenous people of a specific country
- (3) Newly identified but of no use to the man kind
- (4) Novel and non-obvious

175. Which of the following is a fresh water animal that possesses flagellated choanocytes?

- (1) *Euspongia*
- (2) *Meandrina*
- (3) *Gorgonia*
- (4) *Spongilla*

176. Match column I with column II.

	Column I		Column II
a.	Hormone releasing IUD	(i)	Multiload 375
b.	Copper releasing IUD	(ii)	LNG-20
c.	Non-medicated IUD	(iii)	Diaphragm
d.	A barrier contraceptive	(iv)	Lippes loop

Choose the **correct** option.

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(ii), b(iii), c(iv), d(i)
- (3) a(iii), b(iv), c(ii), d(i)
- (4) a(ii), b(i), c(iv), d(iii)

177. Which of the following ARTs involves *in-vivo* fertilisation and will be the most suitable for the couple where the male partner suffers from erectile dysfunction?

- (1) ZIFT
- (2) IUT
- (3) IUI
- (4) ICSI

178. Amniocentesis is used to test for the presence of down syndrome, haemophilia, sickle-cell anemia, etc., as all of them are

- (1) Infectious diseases
- (2) Auto-immune disorders
- (3) Genetic disorders
- (4) Neurodegenerative disorders

179. The pO_2 of oxygenated blood present in the systemic arteries is

- (1) More than the pO_2 in alveoli
- (2) Less than the pCO_2 in deoxygenated blood
- (3) Equal to the pO_2 in tissues
- (4) Less than pO_2 in atmospheric air

180. The factor that does not affect the rate of diffusion of O_2 and CO_2 in alveoli is

- (1) Solubility of these gases
- (2) Thickness of the diffusion membrane
- (3) Concentration gradient of these gases
- (4) Total pressure of the gaseous mixture